

Curriculum Vitae

Goosang Yu
Ph.D.

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EDUCATION

2019.09-2025.02	Ph.D. Medical Science Department of Pharmacology, Yonsei University, College of Medicine, Seoul, Republic of Korea
2017.09-2019.02	M.S. Bioengineering School of Integrative Engineering, Chung-Ang University, Seoul, Republic of Korea
2014.03-2017.08	B.S. Biomedical Engineering (with honors) School of Integrative Engineering, Chung-Ang University, Seoul, Republic of Korea

RESEARCH INTERESTS

Genome editing; high-throughput screening; multi-omics; machine learning and deep learning; bioinformatics

RESEARCH EXPERIENCE

Sanjana Lab Postdoctoral researcher (<i>Advisor: Prof. Neville Sanjana</i>) (2025.10- present)	(New York Genome Center)
Laboratory of Genome Editing Postdoctoral researcher (<i>Advisor: Prof. Hyongbum Henry Kim</i>) (2025.03- 2025.10)	(Department of Pharmacology, Yonsei University College of Medicine)
Laboratory of Genome Editing Ph.D. (<i>Advisor: Prof. Hyongbum Henry Kim</i>) (2019.09- 2025.02)	(Department of Pharmacology, Yonsei University College of Medicine)
Laboratory of Genome Editing Internship (<i>Advisor: Prof. Hyongbum Henry Kim</i>) (2019.03-2019.08)	(Department of Pharmacology, Yonsei University College of Medicine)
Complex Biomaterials Microsystems & Tissue Engineering Lab Integrated Bachelor's and Master's Program (<i>Advisor: Prof. Donghyun Lee</i>) (2017.09-2019.02)	(School of Integrative Engineering, Chung-Ang University)
Complex Biomaterials Microsystems & Tissue Engineering Lab Undergraduate Researcher (<i>Advisor: Prof. Donghyun Lee</i>) (2015.09-2017.08)	(School of Integrative Engineering, Chung-Ang University)

PUBLICATION

* Equal contribution # Corresponding Author

- Jung, Y.*, **Yu, G.***, Oh, H., Lee, J., Jeon, E., Bae, J., Sim, T. & Kim, H. H.# (2025). Comprehensive resistance profiling of ABL1 variants against various kinase inhibitors using prime editing. *Nature Biomedical Engineering*
- Park, J., **Yu, G.**, Seo, S. Y., Yang, J. & Kim, H. H.# (2024). SynDesign: web-based prime editing guide RNA design and evaluation tool for saturation genome editing. *Nucleic Acids Research*, gkae304.
- **Yu, G.***, Kim, H. K.*, Park, J., Kwak, H., Cheong, Y., Kim, D., ... & Kim, H. H.# (2023). Prediction of efficiencies for diverse prime editing systems in multiple cell types. *Cell*, 186(10), 2256-2272.
- Jang, H.*, Jo, D. H.*, Cho, C. S., Shin, J. H., Seo, J. H., **Yu, G.**, ... & Kim, H. H.# (2022). Application of prime editing to the correction of mutations and phenotypes in adult mice with liver and eye diseases. *Nature Biomedical Engineering*, 6(2), 181-194.
- Kim, H. K.*, **Yu, G.***, Park, J., Min, S., Lee, S., Yoon, S., & Kim, H. H.# (2021). Predicting the efficiency of prime editing guide RNAs in human cells. *Nature Biotechnology*, 39(2), 198-206.

SKILLS AND PROFESSIONAL EXPERTISE

- **Molecular experiment techniques**

Nucleic acids extraction	DNA / RNA from mammalian cells or plasmid DNA from bacteria cells
PCR	Extension PCR, Overlap PCR, qPCR
Cloning	Design and cloning plasmid vectors by PNK annealing, T4 ligation, or Gibson assembly
NGS	NGS sampling, sequencer operating

- **Cellular experiment techniques**

Genome editing	CRISPR-Cas9, base editing and prime editing
Transfection	PEI, Lipofectamine, electroporation (Neon or NEPA)
Lentivirus	Stable cell line preparation
Virus-like particle (VLP)	Protein or RNA-protein complex delivery to cells
Screening	High-throughput screening design and experiment in mammalian cells
Differentiation	Osteogenesis, adipogenesis

- **In vivo experiment techniques**

Mouse handling	Maintain, tagging, oral administration
Transplantation	Subcutaneous xenografts

- **Computational skills**

Python	Source code development, multiprocessing, and software deployment
Linux	Setting up and using Ubuntu or CentOS environments
Bioinformatics	Sequencing data analysis and database parsing
Machine learning	Utilizing scikit-learn, PyCaret, PyTorch, and TensorFlow; feature engineering and analysis
MD simulation	Simulation and analysis using GROMACS
PyMol	Protein structure visualization and analysis
Git / GitHub	Source code management and deployment; collaborative program development
Server management	Maintenance of hardware and operating systems

PRESENTATION

- Utilization and optimization of prime editing for functional screening. Genome engineering: CRISPR Frontiers, Cold Spring Harbor Laboratory Meetings & Courses Program, NY, United States. August 2024 [Talk]
- Utilization and optimization of prime editing for functional screening. Developmental Biology Visitor Seminar, Memorial Sloan Kettering Cancer Center, NY, United States. August 2024 [Talk]
- Optimization and Utilization of Prime Editing. Korean Society for Biochemistry and Molecular Biology (KSBMB) International Conference, Seoul, Republic of Korea. May 2024 [Talk and Poster]
- GenET: A python package for genome editing research. Korean Association for Genome Editing (KAGE), Seoul, Republic of Korea. November 2023 [Poster]
- Prediction of efficiencies for diverse prime editing systems in multiple cell types. Genome Engineering Seminar Series (GESS), Harvard Medical School, Boston, United States. October 2023 [Talk]
- Prediction of efficiencies for diverse prime editing systems in multiple cell types. Genome Engineering: CRISPR Frontiers, Cold Spring Harbor, NY, United States. August 2023 [Poster]
- Prediction of efficiencies for diverse prime editing systems in multiple cell types. Biological Research Information Center (BRIC) Webinar, Pohang, Republic of Korea, July 2023 [Talk]
- Evaluation and prediction of the efficiency of prime editor. Korean Society for Biochemistry and Molecular Biology (KSBMB) International Conference, Seoul, Republic of Korea. May 2023. [Poster]
- Evaluation and prediction of the efficiency of prime editor. Frontiers in Genome Engineering (FGE), Seoul, Republic of Korea. July 2022. [Poster]
- Effect of cyclic electrical stimulation to regulate osteogenic differentiation of pre-osteoblast cell line (MC3T3-E1). The Korean Society for Biotechnology and Bioengineering (KSBB), Seoul, Republic of Korea. October 2017. [Poster]

SCHOLARSHIPS and FUNDINGS

2019-2023	Brain Korea 21 Project Graduate Students Support Fund Department of Medical science, Yonsei University, Seoul, Republic of Korea
2023	Academic Research Fellowship Yonsei University, Seoul, Republic of Korea
2021	Graduate student scholarship in biomedical sciences ASAN foundation, Seoul, Republic of Korea
2017-2018	CAU GRS (Chung-Ang University Graduate Research Scholarship) Chung-Ang University, Seoul, Republic of Korea
2015-2017	CK-2 (University of Creative Korea) Scholarship Chung-Ang University, Seoul, Republic of Korea
2014-2017	Da Vinci Scholarship Chung-Ang University, Seoul, Republic of Korea
2014-2015	National Grant Type I Korea Student Aid Foundation, Daegu, Republic of Korea
2014-2015	Department Secondary Honor Scholarship Chung-Ang University, Seoul, Republic of Korea

PATENTS

- **System and method for predicting prime editing efficiency using deep learning.**
Application No. KR102538128B1 (KR), US20230274792A1 (US), CN116508104A (CN), WO2022025623A1 (WO)
- **Method and apparatus for predicting prime editing efficiency of diverse prime editors in various cell types.**
Application No. KR20240034650A (KR), WO2024053921A1 (WO)

HONORS and AWARDS

- Best Presentation Award**, Korean Society for Biochemistry and Molecular Biology, 2024
- Best Academic Paper Award**, Yonsei University, 2024
- ANYGEN Best Research Award**, Korean Society for Biochemistry and Molecular Biology, 2023
- Best Presentation Award**, Department of Medical Science, Yonsei University College of Medicine, 2021
- Merit Academic Paper award**, Yonsei University, 2020
- Excellent poster presentation award**, The Korean Society for Biotechnology and Bioengineering, 2017

TEACHING & WORK EXPERIENCE

2022-2025	Technical Research Personnel (Military Service) Military Manpower Administration, Seoul, Republic of Korea
2019-2023	Mentoring, Internship Program Chronic Intractable Disease Systems Medicine Research Center Mentee: Yoonjin Cho, Jaeh Jeon, Hojin Byun, Dongeun Shin
2022	Graduate Student Instructor Course title: Data Analysis and Research Applications Using Python Yonsei University, Seoul, Republic of Korea
2018	Pilot Teaching Assistant C Chung-Ang University, Seoul, Republic of Korea

Reference

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Professor, Yonsei Lee Youn Jae Fellow

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